CHAPTER- 2 STRUCTURE OF ATOM

Anion - An ion with a net negative charge.

Atomic orbital - An orbital, associated with only one particular atom, in which electrons reside. Though they are called orbitals, atomic orbitals should not be conceived as akin to the orbits of planets rather around a star. Instead, orbitals describe a locus of space in which an electron is likely to reside. Each orbital can hold up to two electrons.

Aufbau principle - German for "building up", a systematic procedure for determining the electron configuration of any atom. Incorporates the Pauli Exclusion Principle and Hund's Rule.

Cation - An ion with a net positive charge.

Degenerate orbitals - Orbitals with identical energies.

Electron - A negatively charged elementary particle of mass $9.109390x10^{-}31$. Electrons of an unbonded atom move around the atomic nucleus in orbitals. Those electrons in the orbitals furthest from the nucleus are the highest in energy, play a crucial role in chemical processes such as bonding, and are called valence electrons.

Electron affinity - The energy change in an atom when it gains an electron.

Electronegativity - A measure of the ability of an atom to attract electrons to itself. Incorporates the atom's ionization energy and electron affinity.

Hund's Rule - A rule which says that, when choosing between orbitals, electrons prefer to go in separate orbitals of the same energy. In this way, every orbital within a particular shell (or subshell when the orbitals are not degenerate) will be ha lf-filled before any single one orbital becomes completely filled.

Ion - Any atom or molecule with a net charge.

Ionization energy - The energy it takes to remove an electron from an atom.

Isoelectronic - Description for two elemental species with the same electronic configuration.

Isotope - Atoms with the same number of protons (i.e. same atomic number) but a different number of neutrons.

Neutron - An uncharged atomic particle of mass $1.67493x10^{-}27$. It resides in the nucleus.

Nucleus - The small, dense central region of an atom around which electrons orbit. The nucleus is made up of protons and neutrons.

Octet rule - The cardinal rule of bonding. The octet rule states that atoms gain stability when they have a full complement of 8 electrons in their valence shells.

Pauli Exclusion Principle - States that no two electrons in an atom or molecule can have the same set of four quantum numbers.

Proton - A positively charged particle of mass $1.6726x10^{-}27$. Protons reside in the nucleus.

Quantum Numbers - The four numbers that define each particular electron of an atom. The Principle Quantum Number (*n*) describes the electrons' energy and distance from the nucleus. The Angular Momentum Quantum Number (*l*) describes the shape of the orbital in which the electron resides. The Magnetic Quantum Number (*m* describes the orientation of the orbital in space. The Spin Quantum number describes whether the spin of the electron is positive or negative.

Shell - A group of subshells of similar energy levels. For example, 2*s* and 2*p*subshells occupy the same shell. Indicated by the principle quantum number.

Shielding - When the attraction from the nucleus felt by one electron is lessened or blocked by intermediate electrons. Shielding can split degenerate orbitals. For example, since *s*-orbital electrons shield for *p*-orbital electrons and receive little shielding themselves, *s*-orbitals are usually of lower energy level than *p*-orbitals of the same shell.

Splitting - Through shielding, the breaking of degenerate orbitals within a shell in multi-electron atoms.

Atoms and Atomic Orbitals

Fundamentals of the Atom

An atom consists of a nucleus of protons and neutrons, surrounded by electrons. Each of the elements in the periodic table is classified according to its atomic number, which is the number of protons in that element's nucleus. Protons have a charge of +1, electrons have a charge of -1, and neutrons have no charge. Neutral atoms have the same number of electrons and protons, but they can have a varying number of neutrons. Within a given element, atoms with different numbers of neutrons are isotopes of that element. Isotopes typically exhibit similar chemical behavior to each other.

Electrons have such little mass that they exhibit properties of both particles and waves; in. We further know from Heisenberg's Uncertainty Principle that it is impossible to know the precise location of an electron. Despite this limitation, there are regions around the atom where the electron has a high probability of being found. Such regions are referred to as atomic orbitals.

Atomic Orbitals and Quantum Numbers

The relation of a particular electron to the nucleus can be described through a series of four numbers, called the Quantum Numbers. The first three of these numbers describe the energy (Principle quantum number), shape (Angular momentum quantum number), and orientation of the orbital (magnetic quantum number). The fourth number represents the "spin" of the electron (spin quantum number). The four quantum numbers are described below.

Principle Quantum Number (*n*)

The principle quantum number indicates how the distance of the orbital from the nucleus. Electrons are farther away for higher values of n . Electrons are negatively charged, so electrons that are closer to the positively charged nucleus are more powerfully attracted and tightly bound than those that are farther away. Electrons that are closer to the nucleus are thus more stable, and less likely to be lost by the atom. In other words, as n increases, so does the energy of the electron and the likelihood of that electron being lost by the atom. In a given atom, all the atomic orbitals with the same n are collectively known as a shell. n can take on integer values of 1 or higher (ex. 1, 2, 3, etc.).

Angular Momentum Quantum Number (*l*)

The angular momentum quantum number describes the shape of the orbital. The angular momentum number (or subshell) can be represented either by a number (any integer from 0 up to n-1) or by a letter (*s, p, d, f, g,* and then up the alphabet), with 0 corresponding to s, 1 to p, 2 to d, and so on. For example:

when $n = 1$, l can only equal 0; meaning that shell $n = 1$ has only an s orbital ($l = 0$).

when $n = 3$, l can equal 0, 1, or 2; meaning that shell $n = 3$ has *s, p,* and *d* orbitals.

s orbitals are spherical, whereas *p* orbitals are dumbbell-shaped. *d* orbitals and beyond are much harder to visually represent.

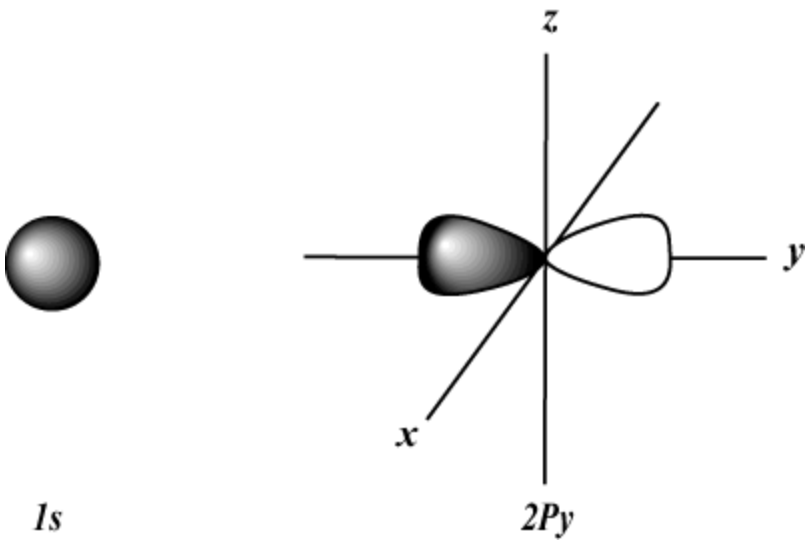


Figure %: s and p atomic orbital shapes

Magnetic Quantum Number (*m*)

Gives the orientation of the orbital in space; in other words, the value of *m*describes whether an orbital lies along the x-, y-, or z-axis on a three-dimensional graph, with the nucleus of the atom at the origin. *m* can take on any value from -*l* to l. For our purposes, it is only important that this quantum number tells us that for each value of *n* there may be up to one *s* -orbital, three *p* -orbitals, five *d* -orbitals, and so on. For example:
The s orbital (*l* = 0) has one orbital, since *m* can only equal 0. That orbital is spherically symmetrical about the nucleus.

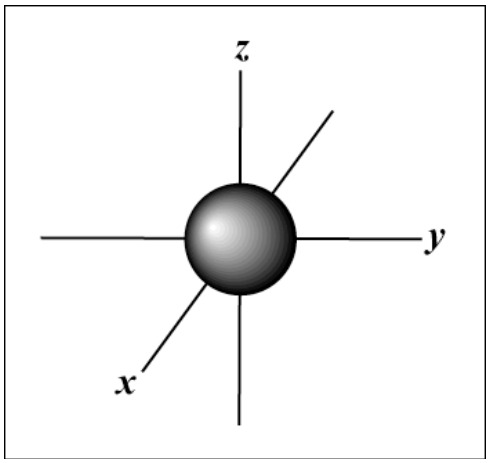


Figure %: s orbital

The *p* orbital (*l* = 1) has three orbitals, since *m* = -1, 0, and 1. These three orbitals lie along the *x* -, *y* -, and *z* -axes.

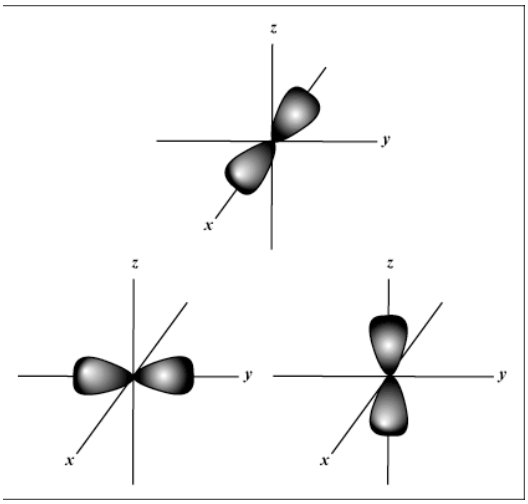
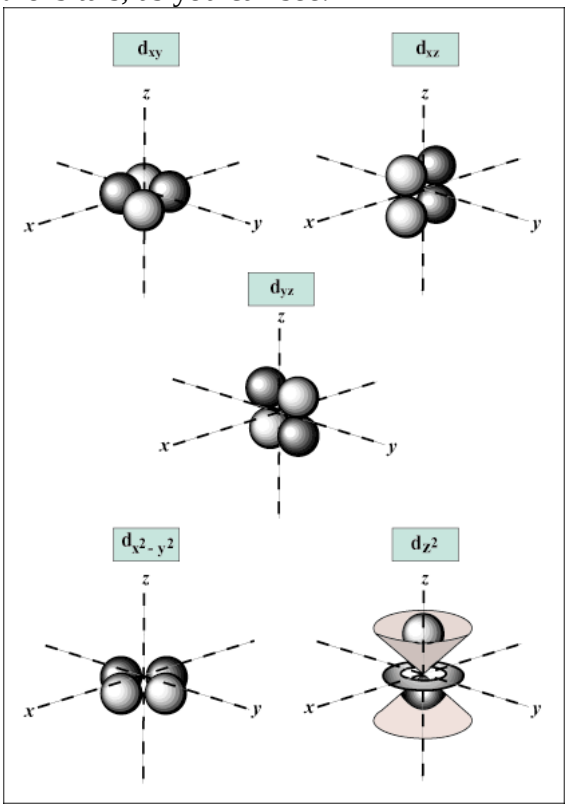


Figure %: p orbitals

The *d* orbital (l = 2) has five orbitals, since m = -2, -1, 0, 1, and 2. It is far more difficult to describe the orientation of *d* orbitals, as you can see:



Subshell - Orbitals of the same subshell are of the same shape and energy. *p*-orbitals are of the same subshell, while *s*-orbitals are of a separate subshell. Indicated by the angular momentum quantum number.

Uncertainty Principle - A tenet of quantum mechanics that says that the position and momentum of any particle cannot both be known precisely at the same time.

Valence electrons - The electrons in the outermost energy shell of an atom. The configuration of these electrons determine the chemical properties of the element.

Valence shell - The highest energy shell in an atom, containing valence electrons. All interactions between atoms take place through the electrons of the valence shell.

Electron Configuration

The electrons in an atom fill up its atomic orbitals according to the Aufbau Principle; "Aufbau," in German, means "building up." The Aufbau Principle, which incorporates the Pauli Exclusion Principle and Hund'sRule prescribes a few simple rules to determine the order in which electrons fill atomic orbitals:

1. Electrons always fill orbitals of lower energy first. 1*s* is filled before 2*s*, and 2*s*before 2*p*.
2. The Pauli Exclusion Principle states no two electrons within a particular atom can have identical quantum numbers. In function, this principle means that if two electrons occupy the same orbital, they must have opposite spin.
3. Hund's Rule states that when an electron joins an atom and has to choose between two or more orbitals of the same energy, the electron will prefer to enter an empty orbital rather than one already occupied. As more electrons are added to the atom, these electrons tend to half-fill orbitals of the same energy before pairing with existing electrons to fill orbitals.

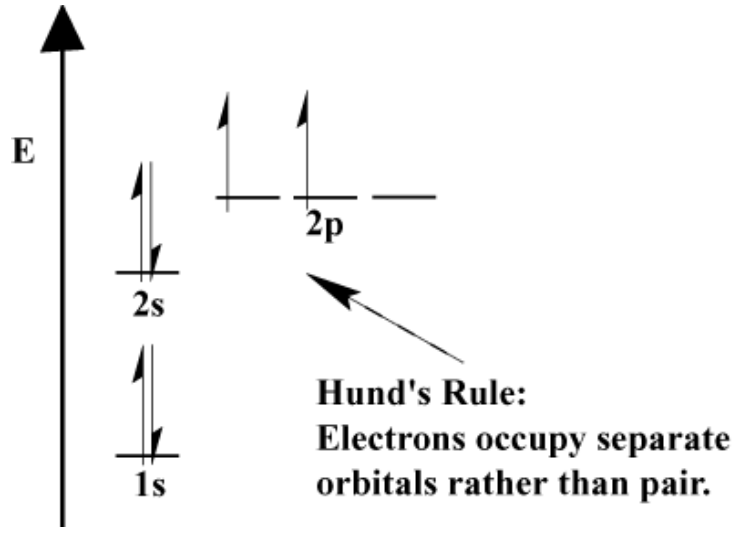


Figure %: The ground state electron configuration of carbon, which has a total of six electrons. The configuration is determined by applying the rules of the Aufbau Principle.

Valency and Valence Electrons

The outermost orbital shell of an atom is called its valence shell, and the electrons in the valence shell are valence electrons. Valence electrons are the highest energy electrons in an atom and are therefore the most reactive. While inner electrons (those not in the valence shell) typically don't participate in chemical bonding and reactions, valence electrons can be gained, lost, or shared to form chemical bonds. For this reason, elements with the same number of valence electrons tend to have similar chemical properties, since they tend to gain, lose, or share valence electrons in the same way. The Periodic Table was designed with this feature in mind. Each element has a number of valence electrons equal to its group number on the Periodic Table.

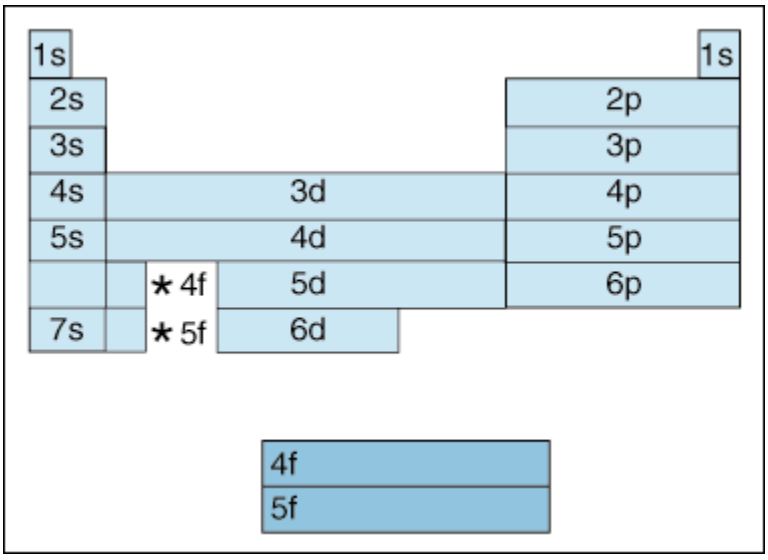


Figure %: The periodicity of valence electrons

This table illustrates a number of interesting, and complicating, features of electron configuration.

First, as electrons become higher in energy, a shift takes place. Up until now, we have said that as the principle quantum number, increases, so does the energy level of the orbital. And, as we stated above in the Aufbau principle, electrons fill lower energy orbitals before filling higher energy orbitals. However, the diagram above clearly shows that the 4*s* orbital is filled before the 3*d* orbital. In other words, once we get to principle quantum number 3, the highest subshells of the lower quantum numbers eclipse in energy the lowest subshells of higher quantum numbers: 3*d* is of higher energy than 4*s*.

Second, the above indicates a method of describing an element according to its electron configuration. As you move from left to right across the periodic table, the above diagram shows the order in which orbitals are filled. If we were the actually break down the above diagram into groups rather than the blocks we have, it would show how exactly how many electrons each element has. For example, the element of hydrogen, located in the uppermost left-hand corner of the periodic table, is described as $1s^1$, with the *s* describing which orbital contains electrons and the ^{1}describing how many electrons reside in that orbital. Lithium, which resides on the periodic table just below hydrogen, would be described as $1s^2 2s^1$. The electron configurations of the first ten elements are shown below (note that the valence electrons are the electron in highest energy shell, not just the electrons in the highest energy subshell).

Atomic Number	Element Symbol	Electron Configuration	Number of Valence Electrons
1	H	$1s^1$	1
2	He	$1s^2$	2
3	Li	$1s^2 2s^1$	1
4	Be	$1s^2 2s^2$	2
5	B	$1s^2 2s^2 2p^1$	3
6	C	$1s^2 2s^2 2p^2$	4
7	N	$1s^2 2s^2 2p^3$	5
8	O	$1s^2 2s^2 2p^4$	6
9	F	$1s^2 2s^2 2p^5$	7
10	Ne	$1s^2 2s^2 2p^6$	8

The Octet Rule

Our discussion of valence electron configurations leads us to one of the cardinal tenets of chemical bonding, the octet rule. The octet rule states that atoms become especially stable when their valence shells gain a full complement of valence electrons. For example, in above, Helium (He) and Neon (Ne) have outer valence shells that are completely filled, so neither has a tendency to gain or lose electrons. Therefore, Helium and Neon, two of the so-called Noble gases, exist in free atomic form and do not usually form chemical bonds with other atoms. Most elements, however, do not have a full outer shell and are too unstable to exist as free atoms. Instead they seek to fill their outer electron shells by forming chemical bonds with other atoms and thereby attain Noble Gas configuration. An element will tend to take the shortest path to achieving Noble Gas configuration, whether that means gaining or losing one electron. For example, sodium (Na), which has a single electron in its outer 3*s* orbital, can lose that electron to attain the electron configuration of neon. Chlorine, with seven valence electrons, can gain one electron to attain the configuration of argon. When two different elements have the same electron configuration, they are called iso-electronic.

VERY SHORT ANSWER TYPE QUESTIONS [1 MARK]

1. How are dxy and $dx^2_-y^2$ orbitals related?

Ans. The dxy orbital is exactly like $dx^2_-y^2$ orbital except that its lobes are at an angle of 45^0 to the lobes of dx^2-y^2 orbital.

2. Calculate the total number of angular nodes present in 3p orbital.

Ans. For 3p orbital, principal quantum number ,n= 3 and azimuthal quantum number , l= 1
Number of angular nodes =l=1.
Number of radial nodes =n-l-1= 3-1-1=1

3. Write the values of the quantum number n,l,m and s for electron filling 21st place in the atom of element with atomic number 24 .

Ans. Electric configuration of element=$1s^2\ 2s^2\ 2p^6\ 3s^2 3p^6 4s^1 3d^5$

21st electron goes to 3d orbital. Its quantum numbers are :n=3, l=2, s=+1/2 or -1/2

M can have any value out of -2,-1,0,+1,+2.since all d-orbitals are degenerate .

$d_{xy} d_{yz} d_{zx}$ $d_{x^2-y^2}$ d_{z^2}

±2 ±1 ±1 ±2 0

4. Which of the following orbitals are degenerate ?

3 d_{xy},4d_{xy} ,3 d_{z^2} ,3 d_{yz}, 4 d_{z^2}

Ans. The orbitals which belongs to same subshell and same shell are called degenerate orbitals.(3 d_{xy} , 3 d_{z^2}, 3 d_{yz}) and (4 d_{xy},4d_{zy}, 4 d_{z^2}) are the two sets of degenerate orbitals.

5. What is the maximum number of electrons that may be present in all the atomic orbitals with principal quantum number 3 and azimuthal quantum number 2?

Ans. Principal quantum number 3 and azimuthal quantum number 2 means 3 dorbitals that can accommodate a maximum of 10 electrons.

6. What is lowest value of n the allows g orbitals to exist?

Sol. For g subshell,l=4 and to have l=4, minimum value of n=5 [because the value of l=0 to(n-1)]

7. An electron is in one of the 3d-ORBITALS. GIVE THE POSSIBLE VALUES OF n,l,m_l FOR THIS ELECTRON.

Ans. For 3d-electron ,n=3,l=0,m=-2,-1,0,+1,+2(anyone)

8. In which atom ,the outermost electron can have the following set of quantum numbers ?

$N=3, l=0, m_l=0, m_s= -1/2$

Ans. N=3 and l=0 means the last electron enters in 3s orbital.thus,the complete configuration upto 3s is

$1s^2 2s^2 2p^6 3s^1$.

Total number of electrons=2+2+6+1=11.thus,the element is sodium.

9. Nickel atom can lose two electrons form ni^{2+}ion.the atomic number of ni is 28.from which orbital will nickel lose two electrons?

ANS. $_{28}Ni = 1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 3d^8, 4s^2$;nickel will lose 2 electrons from 4s(outrmost shell)to form Ni^{2+} ion.

10.show the distribution of electrons in oxygen atom(atomic number 8)using orbital diagram.

ANS.$_8O=1s^1 \quad 2s^2 \qquad\qquad 2p^4$

↿⇂	↿⇂	↿⇂	↿	↿

11.the bromine atom possesses 35 electrons.it contains 6 electrons in 2p orbital, 6 electrons in 3p orbital and 5 electrons in 4p orbital .which of these electron expriences the lowest effective nuclear charge?

ans effective nuclear charge decreases as the distance of the orbitals increases from the nucleus .hence, 4p electrons experience the lowest effective nuclear charge.

12. arranges,p and d-subshells of shell in the incresing order of effective nuclear charge(Z_{EFF})experienced by the electron present in the nucleus.

ANS .s-orbital is spherical in shape it shields the electrons from the nuleus more effectively than p- orbital which in turn shields more effectively THAN d-ORBITAL . THEREFORE, THE EFFECTIVE NUCLEAR CHARGE (Z_{EFF}) EXPERIENCED BY ELECTRONS PRESENT IN THEM IS d<p<s.

Q.12= What is the difference in the origin of cathote rays and anode rays?

Ans. = cathode rays originate from the cathode whereas rays are not obtained from the anode. They are produced from the gaseous atoms by knock out the electrons by high speed cathode rays.

Q.13 = Neutrons can be found in all atomic nuclei except in one case. Which is the atomic nucleus and what does it consist of?

ANS. = in case of hydrogen atom, there is no neutron. It consists of only one person.

Q.14 = Where in an atom is nearly all of its mss concentrated? explain your answer in particles yhat contribute to this mass.

ans. = The whole mass is concentrated in small positive region known as the nucleus. positive charge of the nucleus is due to protons, while the mass of the nucleus is due to the both protons and neutrons.

q.15 = In millikan' s experiment, static electric charge on the oil dropshas been obtained by shining x-rays.

ans. = 8.0 electrons

q.16 = The magnitude of charge on the electrons is 4.8×10^{-10}esu. what is the charge on the helium atom?

ANS. = $+ 9.6 \times 10^{-10}$.

Q.17 = Which of the following does not show the deflection the path on passing through an electric field?

protons, cathode rays, electron, neutron.

ans. = Neutrons is a neutral partical. hence it will not be deflected on passing through an electric field.

q.18 = What is the nuclear radius of an atom whose mass number is 125?

ANS. = $r = Ro \times A^{1/3}$

$r=[1.4 \times 10^{-15}{}_{m}] \times 125^{1/3} = 7.0 \times 10^{-15}$

Q.19= What is the difference between atomic mass and mass number ?

ans .= mass number is whole number because it is the sum of the protons and electron and atomic mass is fractional because it is the average relative mass of its atom as compared with mass of atom of C-12 isotopes taken as 12.

Q.20= An atom having atomic mass number 13 and 7 neutron . what is the atomic number of the atom ?

ANS. = $A = n + p$

$p = A - n = 13 - 7$

atomic number , $Z = p = 6$

Q.21= Name the element whose isotope has mass number 14 and 8 neutrons.

ANS. = ATOMIC NUMBER = MASS NUMBER – NUMBER OF NEUTRONS

ATOMIC NUMBER = 14 – 8 = 6 SO, THE ELEMENT IS CARBON

Q.22= Arrange the following type of radiationx in increasing order of frequency ?

ANS. = The order of frequency is

radituation from fm radio < microwave < amber colour< x-rays < cosmic rays.

Q.23 = How does the intensity of the spectral line vary with wavelength ?

Ans. = As the wavelength decreases . Intensity is also decreases

Q.24 = What is the maximum number of emission lines ehwn the excited electron of a h-atom in n = 6 drops to the ground state ?

ANS. = Number of lines produced when electron from n thshell to ground state = n (n-1)/2

WHEN , n = 6 , NUMBER OF LINES PRODUCED = 6(6-1) / 2 = 6 x 5 / 2 = 15

Q. 25= What do you mean by saying that energy of the electron is quantised ?

ANS. = This means that the electrons is an atom have only definite values of energies

Q.26 = What is the experimental evidence in support pf the idea that electronic energies in an atom are quantised?

ANS . = The line spectrum of any element has lines corresponding to the definite wavelength . Lines are obtained as a result of elrectronic transitions between the energy level .Hence , the electrons in these level have fixed energy .

Q.27= What is the difference between a quantum and a photon ?

ANS . = The smallest packet of ebergy of any radiation is called a quantum whereas that of light is calee photon

Q.28 = Calculate the distance of separation between the second and third orbits of hydrogen atom .

ANS . = For H- atom the radius of nth orbit is given by

$$r_n = 0.529 \times n^2 \text{ Å}$$

$$r_n - r_2 = 0.529 (3^2 - 2^2)$$

$$= 0.529 \times 5$$

$$= 2.645 \text{ Å}$$

Q. 29= Why is the bohr' s orbits called stationary states ?

Ans. = This is beacause the energies of the orbits in which the electrons revolve are fixed .

SHORT ANSWER TYPE QUESTION [2 MARKS]

1. **Chlorophyll present in green leaves absorbs light at 4.620×10^{14} Hz. Calculate the wavelength of radiation in nanometer. Which part of the electromagnetic spectrum does it belong to?**

Sol. $\lambda = \frac{c}{v} = \frac{3.0 \times 10^8 \ ms}{4.620 \times 10^{14} s^{-1}}^{-1} = 649.4$ nm

Thus, it lies in the visible light.

2. **How long would it take a radio wave of frequency, $6 \times 10^3 \ s^{-1}$ to travel from Mars to Earth, a distance of 8×10^7 km?**

Sol. All radiations in vaccum travel with the same speed i.e., $3\times10^{8}\ ms^{-1}$

Distance to be travelled from Mars to the Earth

$= 8 \times 10^{7}$ km $= 8 \times 10^{7} \times 10^{3}$ m (1 km = 10^{3} m)

Time taken = $\frac{8\times10^{7}\times10^{3}}{3\times10^{8}}$

$= 2.66 \times 10^{2}$ s

= 4 min 26 s

3. **The Vividh Bharti section of all India radio, Delhi, broadcasts on a frequency of 1368 kHz (kilohertz). Calculate the wavelength of the electromagnetic radiation emitted by transmitter. Which part of the electromagnetic spectrum does it belong to?**

Sol. Frequency, v = $\frac{c}{\lambda}$

1368×10^{3} Hz = $\frac{3\times10^{8}ms^{-1}}{\lambda}$

$\lambda = \frac{3\times10^{8}ms^{-1}}{1368\times10^{3}}$ Hz

= 219.3 m, radio wave

4. **Yellow light emitted from sodium lamp has a wavelength (λ) of 580 nm. Calculate the frequency (v) and wave number (v) of the yellow light.**

Sol. Frequency, v = $\frac{c}{\lambda}$

1 nm = 10^{-9} m

580 nm = 580×10^{-9} m

580×10^{-7} cm

V = $\frac{3.0\times10^{8}ms^{-1}}{580\times10^{-9}m}$ = $5.17\times10^{-14}\ s^{-1}$

(Velocity of light= $3\times10^{8}\ ms^{-1}$)

Wave number, v= $\frac{1}{\lambda}$= $\frac{1}{580\times10^{-7}cm}$

= 1.724×10^{4} cm^{-1}

5. What is the number of photons of light with a wavelength of 4000 pm that provide 1 J of energy?

Sol. Energy, $\frac{hc}{\lambda}$= $\frac{6.626\times10^{-34}Js\times3.0\times10^{8}ms^{-1}}{4000\times10^{-12}m}$

(1 pm = 10^{-12} m)

=4.9695×10^{-17} J

Number of photons,

N = $\frac{1J}{4.9695\times10^{-17}J}$

= 2.0122×10^{16} photons

6. A 25 watt bulb emits monochromatic yellow light of wavelength 0.57 μm. calculate the rate of emission of quanta per second.

Sol. 25 watt= 25 Js^{-1}

Solve as Q.5

[Ans. Number of photons = 7.169×10^{19} photons]

7. Electromagnetic radiation of wavelength 242 nm is just sufficient to ionize the sodium atom. Calculate ionization energy of sodium in kJ mol^{-1}.

Sol. Energy, E= hv= $\frac{hc}{\lambda}$

= $\frac{6.626\times10^{-34}Js\times3.0\times10^{8}ms^{-1}}{242\times10^{-9}m}$

E = 0.0821×10^{-17} J/ atom

This energy is sufficient for ionization of one Na atom, so it is the ionization energy of Na atom.

E= $6.02\times 10^{23} \times 0.0821\times10^{-17}$ J/atom

E= 4.945×10^{5} J/mol

= 4.945×10^{2} kJ/mol

8. The mass of an electron is 9.1×10^{-31} kg. If its KE (kinetic energy) is 3×10^{-25} J then calculate its velocity.

Sol. KE= $\frac{1}{2}mv^2$

or $v = \frac{2KE^{1/2}}{m} = \frac{2\times3\times10^{-25} kg\ m^2 s^{-2}}{9.1\times10^{-11} kg}^{1/2}$

$= 8.12\times10^{2}$ ms^{-1}

9. What is photoelectric effect? How does particle nature of light explain this effect?

Sol. Refer of page 62.

10. Why the number of lines observed in the hydrogen spectrum is very large?

Sol. Refer to page 68

11. The work function (W_0) of some metals is listed below. Count the number of metals which will show photoelectric effect when light of 300 nm wavelength falls on the metal.

Metal	**Li**	**Na**	**K**	**Mg**	**Cu**	**Ag**	**Fe**	**Pt**	**W**
W_0(eV)	**2.4**	**2.3**	**2.2**	**3.7**	**4.8**	**4.3**	**4.7**	**6.3**	**4.75**

Sol. Given that wavelength is

λ= 300 nm = 3×10^{-7} m

Therefore, energy is

E=hv= $\frac{hc}{\lambda} = \frac{6.6\times10^{-34}\times3\times10^{8}}{3\times10^{-7}\times1.6\times10^{-19}}$ = 4.1 eV

For a metal to show photoelectric effect, its work function has to be less than or equal to 4.1 eV. So the number of metals having work function less than 4.1 eV is 4 i.e., Li, Na, K and Mg.

12. Electrons are emitted with zero velocity from a metal surface when it is exposed to radiation of wavelength 6800 Å. calculate threshold frequency (v_0) and work function (W_0) of the metal.

Sol. Threshold wavelength,

$$\lambda_0 = 6800Å = 6800\times10^{-10}\ m$$

Threshold frequency,

$$v_0 = \frac{c}{\lambda 0} = \frac{3.0\times10^{8} ms^{-1}}{6800\times10^{-10} m} = 4.41\times10^{14}\ s^{-1}$$

Work function, $W_0 = hv_0$

$$= 6.626\times10^{-34} Js \times 4.41\times10^{14}\ s^{-1}$$

$$= 2.922\times10^{-19}\ J$$

13. Calculate the wavelength of the radiation which would cause a photochemical dissociation of a chlorine molecule. The bond dissociation energy of Cl-Cl bond is 245 kJ mol^{-1}.

Sol. Energy required to break one mole of Cl-Cl bonds = 245 kJ

Energy required breaking one Cl-Cl bond

$$= \frac{245\times10^{3}}{6.023\times10^{23}} = 4.068\times10^{-19}\ J$$

Now, $E = hv = \frac{hc}{\lambda}$

Or $\lambda = -\frac{hc}{E} = \frac{6.626\times10^{-34}\times3.0\times10^{8}}{4.068\times10^{-19}} = 4.89\times10^{-7}\ m$

14. Why electronic energy is negative? Comment on the spacing between the energy levels.

Sol. Refer to page 66. The spacing between the energy levels decreases as we move outwards from the nucleus.

15. What transition in the hydrogen spectrum would have the same wavelength as the Balmer transition n=4 to n2 of He^+ spectrum?

Sol. Wave number, $v = RZ^2 (1/ n_1^2 - 1/ n_2^2)$ and $v = \frac{1}{\lambda}$

For He^+ spectrum (for Balmer transition)

$$v = R \times (2)^2 (1/ 2^2 – 1/4^2)$$

$$v = R \times 4 \times \frac{3}{16} = \frac{3}{4}R \qquad \text{…..(i)}$$

For H- spectrum, $v = R \times 1^2 (1/ n_1^2 - 1/ n_2^2)$ …..(ii)

$\frac{3}{4}R = R(1/ n_1^2 - 1/ n_2^2) \Rightarrow \frac{3}{4} = (1/ n_1^2 - 1/ n_2^2)$

Hence, $n_1 = 1$ and $n_2 = 2$. In spectrum, the transition from n= 2 to n= 1 have same wavelength as the Balmer transition from n = 3 to n=2 of He^+ spectrum.

16. What is the wavelength of light emitted when an electron in a hydrogen undergoes transition from an energy level with n=4 to an energy level with n=2?

Sol. $\frac{1}{\lambda} = 109677 (1/2^2 – 1/4^2)\ cm^{-1}$

$\lambda = 486$ nm

The color corresponding to this wavelength is blue.

17. Lifetimes of the molecules in the excited states are often measured by using pulsed radiation source of duration nearly in the nanosecond range. If the radiation source has the duration of 2ns and the number of photons emitted during the pulse source is 2.5×10^{15}, calculate the energy of the source.

Sol. Frequency, v = $\frac{1}{period}$ = $\frac{1}{2ns}$ = $\frac{1}{2\times10^{-9}\ s}$

= $0.5\times10^{9}\ s^{-1}$

Energy of the source = energy of 1 photon × number of photons produced

E_{source} =hv × N

= 6.626×10^{-34}Js × $0.5\times 10^{9}\ s^{-1} \times 2.5\times 10^{15}$

= 8.28×10^{-10} J

18. A gas absorbs a photon of 355 nm and emits at two wavelengths. If one of the emissions is at 680 nm, at what place other is?

Sol. The wavelength of absorbed radiation is related to those of emitted radiation as

$\frac{1}{\lambda absorbed} = \frac{1}{\lambda 1} + \frac{1}{\lambda 2}$

Therefore, $\frac{1}{355} = \frac{1}{680} + \frac{1}{\lambda 2}$

On solving, we get λ_2 = 743 nm

19. The ionization energy of H- atom (in the ground state) is x kg. Find the energy required for an electron to jump from second to third energy level.

Sol. Energy in the second level, $E_2 = -\frac{x}{2^2} = -\frac{x}{4}$

Energy in third energy level,

$$E_3 = -\frac{x}{3^2} = -\frac{x}{9}$$

$$E_3 - E_2 = -\frac{x}{9} + \frac{x}{4} = \frac{5x}{36}$$

20. Calculate the energy required for the process

$$\mathbf{He^{+}_{(g)} \rightarrow He^{2+}_{(g)} + e^{-}}$$

The ionization energy for the H- atom in the ground state is 2.18×10^{-18} J $atom^{-1}$.

Sol. Energy of electron in unielectron atomic system,

$$E_n = \frac{-2\pi^2 m Z^2 e^4}{n^2 h^2}$$

For H-atom, ionization energy (IE) $= E_\infty - E_1$

$$IE = 0 - \frac{-2\pi^2 m e^4 1^2}{1^2 h^2}$$

(Where Z=1 and n=1 for H-atom)

$IE = 2.18\times10^{-18}$ J atom^{-1}.

For He^+, IE $= E_\infty - E_1 = 0 - \frac{-2\pi^2 m e^2 2^2}{1^2 h^2}$

$= 4\times \frac{2\pi^2 m e^4}{h^2} = 4\times2.18\times10^{-18}$ J atom^{-1}.

$= 8.72\times10^{-18}$ J atom^{-1}

The energy required for the process

$He^+_{(g)} \rightarrow He^{2+}_{(g)} + e^-$ is 8.72×10^{-18} J atom^{-1}

21. (i) The energy associated with the first orbit in the hydrogen atom is -2.8×10^{-18} J atom^{-1}. What is the energy associated with the fifth orbit?

(ii) Calculate the radius of Bohr's fifth orbit for hydrogen atom.

Sol. (i) Energy in n^{th} orbit, $E_n = \frac{-2.8\times10-18}{n^2}$ J

Energy in fifth orbit,

$E_5 = \frac{-2.8\times10-18}{5^2}$ J $= -8.72\times10^{-20}$ J

(ii) For H-atom, radius of nth orbit, $r_n = 0.529\times n^2$ Å

Radius of 5th Bohr orbit,

$r_5 = 0.529\times5^5 = 1.3225$ nm

22. For hydrogen like particle, derive the expression $v_n = (Ze^2/mr_n)^{1/2}$ where v_n is the velocity of the electron at distance r_n from the nucleus, Z is the atomic number of the H-like particle, m and e are the charge and mass of the electron.

Sol. For H-like particle, force of attraction between the electron and the nucleus = centrifugal force

i.e., $\frac{Ze^2}{r^2n} = \frac{mvn2}{rn}$

$$v_n^2 = \frac{Ze^2}{mvn}$$

$v_n = \left(\frac{Ze^2}{mvn}\right)^{1/2}$

23. In astronomical observations, signals observed from the distant stars are generally weak. If the photon detector receives a total of 3.15×10^{-18} J from the radiations of 600 nm, calculate the number of photons received by the detector.

Sol. Solve just like Q.5

[Ans. 10 photons]

24. Nitrogen laser produces a radiation at a wavelength of 337.1 nm. If the number of photons emitted is 5.6×10^{24} , calculate the power of this laser.

Sol. Energy of 1 photon, $E = \frac{hc}{\lambda}$

$= \frac{6.626\times10^{-34}\times3.0\times10^{8}}{337.1\times10^{-9}m}$ (1 nm = 10^{-9})

$= 0.05896 \times10^{-17}$ J

Energy of 5.6×10^{24} photons

$= 0.05896 \times10^{-17}$ J $\times$ 5.6×10^{24} J

$= 0.3302 \times 10^{7} = 3.302\times10^{6}$

25. Discuss the similarities and differences between the 1s and 2s orbitals.

Sol. The similarities in 1s and 2s orbitals are as follows

(i) Both have spherical shape.

(ii) Both have the same angular momentum.

The differences in between them are as follows

(i) Size of 2s is larger than that of 1s.

(ii) Energy of 2s is greater than that of 1s.

(iii) 2s has one node while is has no node.

26. (a) How many subshells are associated with n=4?

(b) How many electrons will be present in the subshell having, m_s value of -1/2 for n=4?

Sol. (a) n=4, l= 0,1,2,3, i.e., s p d f (4 subshells)

(b) Number of orbitals in 4th shell = $n^2 = 4^2 = 16$

The maximum number of electrons present in any orbital is two and each orbital has one electron with $m_{s=}$ - ½. Hence, there are 16 electrons with $m_{s=}$ - ½.

27. Using s p d notations, describe the orbital with the following quantum numbers.

(i) n=1, l=0 **(ii) n=3, l=1**

(ii) n=4, l=3. **(iv) n=4, l=3**

Sol.	n	l	sudshell notation
(i)	1	0	1s
(ii)	3	1	3p

(iii) 4 2 4d

(iv) 4 3 4f

28. Which orbital in each of the following pairs is lower in energy in a many-electron atom?

(i) 2s, 2p

(ii) 3p, 3d

(iii) 3s, 4s

(iv) 4d, 5f

Sol. (i) 2s<2p (ii) 3p<3d

(iii) 3s<4s (iv) 4d<5f

This is because lower value of (n+l), lower is the energy and if (n+l) are same, the orbital with lower value of n is of lower energy.

29. Hydrogen atom has only one electron, so mutual repulsion between electrons is absent. However, in multielectron atom mutual repulsion between the electrons is significant. How does this affect the energy of an electron in the orbitals of the same principal quantum number in multielectron atom?

Sol. In hydrogen atom, the energy of an electron is determined by the value of n and in multielectron atom, it is determined by n+l. hence, for a given principal quantum, electrons of s, p, d and f-orbitals have different energy (for s, p, d and f, l=0,1,2 and 3 respectively).

30. An atom of an element contains 29 electrons and 35 neutrons. Deduce

(i) The number of photons

(ii) **The electronic configuration of the element**.

Sol. For a neutral atom,

Number of electrons= number of protons

29 electrons= 29 protons

$_{29}Z = 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}\ 4s^1$

(The element is copper)

31. write the electronic configuration of $_9F^{19}$ $_{16}S^{32}$ and $_{18}Ar^{38}$ and then point out the element with

(i) maximum nuclear charge

(ii) minimum number of neutrons

(iii) **Maximum number of unpaired electrons**.

Sol. $_9F^{19} = 1s^2\ 2s^2\ 2p_x^2\ 2p_y^2\ 2p_z^1$

$_{16}S^{32} = 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p_x^2\ 3p_y^1\ 3p_z^1$

$_{18}Ar^{38} = 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$

(i) maximum nuclear charge = 18 in $_{18}Ar^{38}$

(ii) minimum number of neutrons = 10 in $_9F^{19}$

(iii) maximum number of unpaired electrons= 2 in $_{16}S^{32}$

32. The unpaired electrons in Al and Si are present in 3p-orbital. Which electrons will experience more effective nuclear charge from the nucleus?

Sol. $_{13}Al = 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^1$

$_{14}Si = 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^2$

Si (+4) has greater nuclear charge than aluminium (+3). Hence, 3p unpaired electrons of Si experience greater effective nuclear charge than Al.

33. Out of Cu^{2+}, Fe^{2+} and Cr^{3+}, which ion is most paramagnetic and why?

Sol. Electronic configuration of

$Cu^{2+} = [Ar]\ 3d^9$

$Fe^{2+} = [Ar]\ 3d^6$

$Cr^{3+} = [Ar]\ 3d^3$

Paramagnetic depends on the number of unpaired electrons. From the electronic configuration of these ions, it is clear that Cu^{2+} has one, Fe^{2+} has four and Cr^{3+} has three unpaired electrons.

34. The electronic configuration of valance shell of $3d^{10}\ 4s^1$ and $3d^9\ 4s^2$. How is this configuration explained?

Sol. Configurations with fully filled and half- filled orbitals have extra stability. In $3d^{10}4s^1$, d-orbital are completely filled and s-orbital is half-filled. Hence, it is more stable configuration.

35. In each of the following pairs of salts, which one is more stable?

(i) Ferrous and ferric salts

(ii) Cuprous and cupric salts

Sol. (i) in ferrous salts Fe^{2+}, the configuration is $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^6$. In ferric salts Fe^{3+}, the configuration is $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^5$. As half-filled $3d^5$ configuration is more stable, ferric salts are more stable than ferrous salts.

(iii) In cuprous salts the electronic configuration of Cu^+ is $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}$. In cupric salts, the configuration of Cu^{2+} is $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^9$. Although Cu^+

has completely filled d orbital, yet cuprous salts are less stable. This is because the nuclear charge is not sufficient enough to hold 18 electrons of Cu^+ ion present in outermost shell.

66.Why does the charge to mass ratio of positive rays depend on the residual gas in the discharge tube? Why is the charge to mass ratio of all cathode rays the same?

Sol. In case of positive rays, the ions remaining after the loss of electrons might have the same magnitude of charge, but different masses. Hence have different charge to mass ratio. Cathode rays are made up of electrons and all electrons have same charge to mass ratio. That's why charge to mass ratio of all cathode rays is same

37. Compute the properties of electrons, protons and neutrons.

Sol. Proton :-it is positively charged particle of atomic nucleus. The atomic number of an element represents the number of of protons in the atom

Electron:- it is negatively charge particle that can occupy a volume or space around an atomic nucleus

Neutrons:- it is unchanged particle of nucleus of all toms except hydrogen. For a given element the mass number is the number of protons and neutrons in the nucleus.

38. Briefly describe the experiment that led to discovery of proton

Sol. Rutherford gold foil experiment, Rutherford bombarded a beam of alpha particles on an ultra thin gold foil and then detected the scattered alpha particles in zinc sulphide screen

Results :-1. Most of particles pass through the foil without any deflection

2.:- some of the particles deflect at small angles

3:- very few even bounce back

39. If the diameter of a carbon atom is 0.15 nm calculate the number atoms which a can be placed by side by side in as straight line across length of scale of length 20 cm long.

Sol. Diameter of a carbon atom =0.15 nm = 0.15×10^{-9}m

Length along which atoms are to be placed =20 cm =0.2 m

Number of carbon atom which can be placed in the given length

$0.2 \div 0.15 \times 10^{-9}$ m=1.33×10^{9} atoms

40. Write down the quantum numbers n and l for the following orbitals

a. 2p b.3d c. 5f

Ans. a. n=2, l= 1 b. n= 3, l=2 c. n= 5, l=3

41. The diameter of zinc atom is 2.6 Ä. Calculate (i) radius of zinc atom in pm and (ii) number of atoms present in length of 1.6cm if the zinc atoms are arranged side by side length wise.

Sol. (i) diameter of zinc = 2.6 Ä = 2.6×10^{-10}m

Radius of zinc atom =2.6×10^{-10} m/2 =130pm

(ii) given , length =1.6cm=1.6×10^{-2}m

Number of zinc atom in 1.6×10^{2} m

1.6×10^{-2}m/2.6×10^{-10}=6.154×10^{7} atoms

42. Define atomic number and mass number.

Sol. Atomic number refers to the number of protons present in the nucleus of an element.

The mass number is the average of protons and neutrons in the nucleus of all existing isotopes of an element.

43. Symbol$^{79}_{35}$Br and ^{79}Br can be written, whereas symbol $^{35}_{79}$br and ^{35}br are not acceptable. Answer briefly.

Sol. The composition of any atom can be represented by using the normal element symbol (X) with superscript on the left hand side as the mass number(A) and subscript (Z) on the left side as the atomic numbered., $^{A}_{Z}X$. Hence, the symbol$^{35}_{79}$BR and ^{35}Br are not acceptable.

NOTE- Atomic number is always written as subscript and mass number as superscript. However, both of thiscan be written on left or right side of the symbol.

44. calculate the approximately charge in the coulomb and approximately mass in kilogram of the nucleus of the lithium 7 isotope

Sol. Nucleus of Li atom has 3 protons and 4 neutrons. Charge on one proton=1.60×10^{-19} coulomb

Charge on 3 protons =$3\times1.60\times10^{-19}C$

=$4.80\times10^{-19}C$

Mass of proton≈mass of neutron≈$1.67\times10^{-27}kg$

Mass of nucleus =$7\times1.67\times10^{-27}$

$11.69\times10^{-27}kg$

45.give the number of electrons in the species H_2^+ ,H_2 and O_2^+

Sol. $H_2={}_1H + {}_1H =1+1= 2_e^-$

$H_2=2-1=1e^-$

$O_2={}_8O={}_8O=8+8=16_e^-$

$O_2=16-1=15_e^-$

46. In Rutherford experiment generally the thin foil of Type equation here. **heavy atoms like gold platinum etc. have been used to be bombarded by ∝ particles. If the thin foil of light atoms like aluminium, etc. is used what is the difference would be observed from the above results?**

Heavy atoms such as gold platinum have heavy nucleus heavy nucleus contains large amount of positive charge. When a beam of $\propto particles$ is shot at athin gold foil most of them pass through without much effect.

Some however are the deflected back or by small angles due to enormous repulsive force of heavy nucleus .if the light aluminum foil is used the number of $\propto particles$ deflected back or those deflected by small angles will be negligible.

47.How does the Bohr theory of the hydrogen atom differ that of Schrodinger?

Sol- Bohr's theory does not consider the de-Broglie concept of dual nature of electron and also contradicts with the Heisenberg's uncertainty principle,while the Schrodinger equation is based on quantum mechanics which deal with the microscopic obejcts having both the particle as well as wave like character.

48.What kind of information about an electron in an atom is obtained from its wave function?

Sol- The square of the amplitude of the electron wave, i.e, ω^2at any point gives probability of finding an electron that point. Since,the region around the nucleus which represents the electron density at different points is called an orbital, hence the wave function for an electron in an atom is called orbital wave function.

49. Calculate the wavelength of an electron moving the velocity of 2.05×10^7 ms^{-1} .

Sol. We know that mass of an electron,

$$M=9.11\times10^{-31}kg,$$

Applying, $\lambda = h/mv = 3.5\times10^{-11}m$

50.Dual behavior of matter proposed by de-Broglie led to the discovery of electron microscope often used for the highly magnified images of biological molecules and other type of materials.If the velocity of the electron in this microscope is 1.6×10^6 ms^{-1},calculate de-Brogile wavelength with electron.

Sol- Using de-Broglie wavelength, $\lambda=h/mv$ we get

$$\lambda=455 \text{ pm}$$

51. Similar to electron diffraction, neutron diffraction microscope is also used for the determination of the structure of molecules. If the wavelength used here is 800 pm, calculate the characteristics velocity associated with the neutron (mass of neutron=1.675×10^{-27}kg).

Sol- Wavelength , $\lambda=h/mv$

Or

$$V= 6.626\times10^{-34}\text{kg m}^2\text{s}^{-1}/ 1.675\times10^{-27}\text{kg}\times800\times10^{-12}\text{m}$$

$$=0.494\times10^{3}\text{ms}^{-1} =4.94\times10^{2}\ \text{ms}^{-1}$$

52. A proton is moving with kinetic energy 5×10^{-27}J. What is the velocity of the proton?

Sol- Mass of proton = $1.008\times10^{-3}/6.02\times10^{23}$kg

$$= 1.67\times10^{-27}\text{ kg}$$

$$KE=1/2\ mv^2$$

Or $$v = 2KE/M =2\times5\times10^{-27}/1.67\times10^{-27} = 5.98$$

53. In an atom, an electron is moving with a speed of 600 ms^{-1} with an accuracy of 0.005% .Find the certainty with which the position of the electron can be located.($h=6.6\times10^{-34}$ kg m^2s^{-1},mass of electron $m_e=9.1\times10^{-31}$kg).

Sol- Using Heisenberg's uncertainty principle,

$$\Delta v=600\times0.005/100=0.3\ ms^{-1}$$

We get, $$\Delta x= 6.6/4\times3.14\times9.1\times10^{-31}\times0.03$$

$$=1.92\times10^{-3}\text{m}$$

54. The velocity associated with a proton moving in a potential difference of 1000 V is 4.37×10^5 ms^{-1} .If the hockey ball of mass 0.1 kg is moving with this velocity , calculate the wavelength associated with this velocity.

Sol- Wavelength associated with the velocity of hockey ball

$$\lambda = h/mv= 6.626\times10^{-34}\text{kgm}^2\text{s}^{-1}/0.1\text{ kg}\times4.37\times10^{5}\ \text{ms}^{-1}$$

$$=15.16\times10^{-39}{}_{\text{m}} =1.516\times10^{-38}\ \text{m}$$

55.Prove that if the uncertainty in position of a moving electron is equal to its de-Brogile wave length then its velocity is completely uncertain.

Sol- Let the uncertainty in position beΔx.

$\Delta x=\lambda$ (de-Broglie wavelength)

Using de-broglie relationship, λ = hlmv.

Putting $\lambda = \Delta x$, we get

$\Delta x=h/mv=h/p$

According to Heisenberg's uncertainty principle

$\Delta x.\Delta P \approx h/4Л$

$h/p\times\Delta p\approx h/4Л$ or $\Delta P/P\approx 1/4Л$

Now, $P=m\times v$

And $\Delta P=m\times\Delta v$, so

$\Delta v/v\approx 1/4Л \Rightarrow \Delta v\approx v/4Л$

thus, uncertainty in velocity is so large that its velocity is uncertainty.

56.the effect of uncertainty principle is significant only for motion of microscopic particles and is negligible for the macroscopic particles justify the statement with the help of suitable example.

Sol. If uncertainty principle is applied to an object of mass say about (10^{-6})then

$\Delta v.\Delta x = h/4Лm$

$\Delta v.\Delta x = 6.626\times10^{-34}$ kg $m^2{}_s{}^{-1}/4\times3.14\times10^{-6}$ kg

(iv) $=0.52\times10^{-28}$ $m^2{}_s{}^{-1}$

(v) The value of $\Delta v\Delta x$ obtained is extremely small and is insignificant. Therefore for milligram sized or heavier objects the associated uncertainties are hardly of any real consequence.

[3 marks question]

1. State (a)Hund's Rule of maximum Multiplicity (b) Aufbau Principle (c) n+l rule

Ans.(a) Pairing of electrons in the orbitals belonging to the same subshell (p, d or f) does not takeplace until each orbital belonging to that subshell has got one electron each i.e., it is singly occupied.

(b)In the ground state of the atoms, the orbitals are filled inorder of their increasing energies

(c) Orbitals with lower value of (n+l) have lower energy. If two orbitals have the same value of (n+l) then orbital with lower value of n will have lower energy.

2. 2×10^8 atoms of carbon are arranged side by side. Calculate the radius of carbon atom if the length of this arrangement is 2.4 cm

Sol total length =2.4 cm

Number of carbon along the length = 2×10^8

Diameter of 1 carbon

$=2.4\text{ cm}/2\times10^8 = 1.2\times10^{-8}$

Radius of 1 carbon atom $=1.2\times10^{-8\text{ cm}}/2\times10^8$

$=0.060$cm

3. Write the 3 points of difference between orbit and orbital.

Ans: Orbit

1. An orbit is a well defined circular path around the nucleus in which the electrons revolve

2. It represents the planar motion of an electron around the nucleus

3. All orbits are circular and disc like.

Orbital

1. An orbital is the three dimensional space around the nucleus within which the probability of finding an electron is maximum(upto 90 %)

2. It represents the three dimensional motion of an electron around the nucleus

3. Different orbitals have different shapes, i.e., s-orbitals are spherically symmetrical, p-orbitals are dumb-bell shaped and so on.

4. State Heisenberg's uncertainty principle. Calculate the uncertainty in the position of an electron if the uncertainty in its velocity is $5x10^5$m/s.

Ans. It states that it is impossible to determine simultaneously, the exact position and exact momentum (or velocity) of an electron. The product of their uncertainties is always equal to or greater than h/4π. Δx x (m x Δv) = h/4π

Δx = h/4πx m x Δv = $6.6x^{-34}$/4 x 3.14 x 9.1 x 10^{-31} X5$x10^{5}$ = 1.0 x 10^{10}

5. Write 3 points of differences between electromagnetic waves and matterwaves.

Electromagnetic waves

1. These are associated with electric and magnetic fields

2. They do not require any medium for propagation.

3. They travel with the same speed a that of light

Matter waves

1. These are not associated with

electric and magnetic field.

2. They require medium for Propagation

3. They travel with lower speeds not constanr for all matter wave

6. (i) Calculate the number of electrons which will together weigh one gram.

(ii) Calculate the mass and charge of one mole of electrons

Ans. (i) Mass of one electron = 9.10939 ×10^{-31}kg

Number of electrons that will weigh 1 g = (1 × 10^{-3})kg

= 0.1098 × 10^{-3+10}

= 0.1098 ×10^{28} = 1.098 ×10^{27}

(ii) Mass of one electron= 9.10939 ×10^{-31}kg

Mass of one mole of electron = 9.10939 ×10^{-31}kg X 6.022 ×10^{23}

= 5.48 ×10^{-7}

Charge on one electron = 1.6022 × 10^{-19} coulomb

Charge on one mole of electron =1.6022 × 10^{-19} coulomb X 6.022 ×10^{23}

= 9.65 × 10^{4} C

Q7. Prove that density of the nucleus is constant.

Density of nucleus = mass/volume = $\frac{A\times1.66\times10^{-27}\ kg}{{}^{4}/_{3}\times(1.33\times10^{-11}\times A^{1/3}}$

$$=\frac{1.66\times10^{-27}}{4\times3.14\times(1.33\times10^{-11})^{3}}\ kg/m^{3}$$

= constant

Thus, density of nucleus is constant, independent of the element under consideration.

Q8.Write down the quantum number n, l and m for the following orbitals

(i)$3d_{x^2-y^2}$ (ii)$4d_{z^2}$

(iii)$3d_{xy}$(iv)$4d_{xz}$

Ans(i)n=3,l=2,m=+2

(ii)n=4,l=2,m=0

(iii)n=3,l=2,m=-2

(iv)n=4,l=2,m=+1.

9. Calculate the approximately charge in the coulomb and approximately mass in kilogram of the nucleus of the lithium 7 isotope

Sol. Nucleus of Li atom has 3 protons and 4 neutrons. Charge on one proton=1.60×10^{-19} coulomb

Charge on 3 protons =$3\times1.60\times10^{-19}$C

=4.80×10^{-19}C

Mass of proton≈mass of neutron≈1.67×10^{-27}kg

Mass of nucleus =$7\times1.67\times10^{-27}$

11.69×10^{-27}kg

10.Prove that if the uncertainty in position of a moving electron is equal to its de-Brogile wave length then its velocity is completely uncertain.

Sol- Let the uncertainty in position beΔx.

$\Delta x=\lambda$ (de-Broglie wavelength)

Using de-broglie relationship, λ = hlmv.

Putting $\lambda = \Delta x$, we get

$\Delta x = h/mv = h/p$

According to Heisenberg's uncertainty principle

$\Delta x.\Delta P \approx h/4\pi$

$h/p \times \Delta p \approx h/4\pi$ or $\Delta P/P \approx 1/4\pi$

Now, $P = m \times v$

And $\Delta P = m \times \Delta v$, so

$\Delta v/v \approx 1/4\pi \Rightarrow \Delta v \approx v/4\pi$

Thus, uncertainty in velocity is so large that its velocity is uncertainty.

10. Difference between photon and quantum

Photon vs Quantum

Photon is an elementary particle. Quantum is considered to be a discrete packet with energy stored in. Photon and quantum are two very important concepts discussed in modern physics. These concepts are widely used in fields such as quantum physics, quantum chemistry, electromagnetic theory, optics, particle physics etc. These concepts are also applied in many real world applications such as LASERs, high resolution microscopy, measurements of molecular distances, quantum cryptography, and photochemistry. It is vital to have a good understanding in photon and quantum in order to excel in such fields. In this article, we are going to discuss what photon and quantum are, their definitions, applications of photon and quantum, the similarities between them and finally the difference between photon and quantum.

Q11. An element with mass number 81 contains 31.7% more neutrons as compared to protons. Assign the atomic symbol.

Ans:- We know that mass of the element,

$$A = p + n = 81$$

Let the no. of proton, $p+ x$

Then, no. of neutrons,

$$n = x + 31.7x/100 = 1.317x$$

(As no. of neutrons are 31.7% more than protons)

Hence, from Eq. (i)

$$x + 1.317x = 81$$

$$2.317x = 81$$

Or $x = 81/2.317 = 34.985 = 35$ (approx.)

Therefore, no. of protons = 35 and the symbol is Br.

(Number of protons = Atomic number)

Q12. Describe Milikan experiment to determine the charge on an electron.

Ans:-In this method, oil droplets in the form of mist, produced by the atomizer, were allowed to enter through a tiny hole in the upper plate of electrical condenser. The downward motion of these droplets was viewed through the telescope, equipped with a micrometer eye piece. Milikan

Concluded that the magnitude of electrical charge q on the droplet is always an integral multiple of the electrical charge e, that is, q=ne, where n=1,2,3…..

Q13. An ion with mass no. 56 contains 3 units of positive charge and 30.4% more neutrons than the electrons. Assign the symbol to this ion.

Ans:-Let the no. of neutrons in an ion = x

Therefore, No. of neutron = x +30.4x/100 = 1.304x.

(no. of neutron are 30.4% more than the no. of neutron)

In the neutral atom, no. of electrons = x + 3

(the ion carries +3 charge)

So, no. of protons = x +3

We know that, the mass no. = n + p

$$=1.304x + x + 3 = 56$$

Or 2.304x = 53, x=53/2.304 =23.003 ~23

Therefore, no. of proton = 23+3 = 26 = atomic no.

Therefore, the symbol of the ion is Fe^{3+}.

Q14. Prove that density of the nucleus is constant.

Density of nucleus = mass/volume = $\frac{A\times1.66\times10^{-27}\ kg}{{}^{4}/_{3}\times(1.33\times10^{-11}\times A^{1/3}}$

$$=\frac{1.66\times10^{-27}}{4\times3.14\times(1.33\times10^{-11})^{3}}\ kg/m^{3}$$

= constant

Thus, density of nucleus is constant, independent of the element under consideration.

Q15. Write the complete symbol for the atom with the given atomic number (Z) and the atomic mass (A).

(i)Z = 17, A = 35 (ii)Z = 92,A = 233

(iii)Z = 4, A = 9

Ans:- (i)$^{35}_{17}Cl$ (ii)$^{233}_{92}U$ (iii)$^{9}_{4}Be$

Q16. According to Rutherford's model of the atom, where is the +ve charge concentrated? What does it predict about the relative size of the nucleus and the radius of the atom? What is the nucleus composed of according to his model?

Ans:- According to Rutherford's model of an atom, the +ve charge is concentrated at the center, that is, at the nucleus. Since, a small fraction of α - particle were deflected through small angle and also a few of them bounced back, it proved that the nucleus occupies a very small volume in the atom. The radius of the atom is about 10^{-10} m while of nucleus is 10^{-15} m. The nucleus is composed of protons and neutrons and is surrounded by revolving electrons.

Q17. Which of the following are Isoelectronic species, i.e., these having the same no. of electrons? $\mathbf{Na^{+}, K^{+}, Mg^{2+}, Ca^{2+}, S^{2-}}$

Ans:- Isoelectronic species have the same no. of electrons but different atomic numbers.(Number of positive charge shows the no. of electrons lost and number of negative charges shows number of electrons gained by an atom. Calculation of no. of electrons have been shown below :-

$$Na^{+} = 11 - 1 = 10e^{-}$$

$$Mg^{+} = 21 - 2 = 10e^{-}$$

$$K^{+} = 19 - 1 = 18e^{-}$$

$$Ca^{2+} = 20 - 2 = 18e^{-}$$

$$S^{2-} = 16 + 2 = 18e^{-}$$

Q18. Find the no. and mass of neutrons in 32mg of O_2 at STP (mass of a neutron = 1.675×10^{-27}kg).

Ans:- 1mole of O_2= 16g = $(6.022\times 10^{23}) \times (8+8)$neutrons

$= 9.6352 \times 10^{24}$neutrons

Now, 16g ofO_2 = 1mol

Therefore, 32g of O_2 = 2mol

Number of neutrons in 32mg ofO_2

$=9.6352\times 10^{24} \times 2 \times 10^{-3}$

$=1.9270\times 10^{22}$

Mass of 1 neutron = 1.675×10^{-27}kg

Therefore, Mass of 1.9270×10^{22}neutrons

$=1.9270 \times 10^{22} \times 1.675 \times 10^{-27}$

$=3.228 \times 10^{-5}$kg

Q19. Which of the following sets of orbitals are degenerate and why?

(i)1s, 2s and 3s in Mg atom.

(ii)$2p_x$, $2p_y$and $2p_z$ in C atom

(iii)3s,$3p_x$ and 3d orbitals in H atom.

Ans:-

(i)1s,2s and 3s orbitals in Mg atom are not degenerate because these have different values of n.

(ii)$2p_x$, $2p_y$and $2p_z$orbitals in C atom are degenerated because these belongs the same subshell.

(iii)3s,$3p_x$ and 3d orbitals in H atom are degenerated because for H atom, the subshells having same value of n have same energies.

Q20. (i)What is the common between $d_{xy}$and$d_{x^2-y^2}$orbitals ?
(ii)What is the difference between them?
(iii)What is the angle between the lobes of the above two orbitals?

Ans:-(i)Both have identical shape, consisting of four lobes.
(ii)Lobes of $d_{x^2-y^2}$lie along the x and y-axis whereas those of d_{xy}lie between x and y-axis.
(iii)45°

Q21.Explain, giving reason which of the following sets of quantum numbers are not possible?

(i)n=0,l=0,$m_l = 0, m_s = +\frac{1}{2}$

(ii)n=1,l=0,$m_l = 0, m_s = -\frac{1}{2}$

(iii)n=1,l=1,$m_l = 0, m_s = +\frac{1}{2}$

(iv)n=2,l=1,$m_l = 0, m_s = -\frac{1}{2}$

(v)n=3,l=3,$m_l = -3, m_s = +\frac{1}{2}$

Ans:-(i)is not possible as n≠0.

(ii)is possible.

(iii)is not possible because if n=1,l=0 only(l≠1)

(iv)is possible.

(v)is not possible because if n=3,l=0,1and 2(l≠3)

Q22.Correct the following electronic configuration of the elements in the ground state.

(i)$1s^2, 2s^2, 2p_x^2, 2p_y^2, 2p_z^2, 3s^2, 2p_x^1$.

(ii)$1s^2, 2s^2, 2p_x^2, 2p_y^2, 2p_z^1$.

(iii)$1s^2, 2s^1, 2p^6, 3s^2, 3p^6, 3d^5$.

(iv)$1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 3d^4, 4s^2$

Ans(i)$1s^2, 2s^2, 2p_x^2, 2p_y^2, 2p_z^2, 3s^2$.

(ii)$1s^2, 2s^1, 2p_x^1, 2p_y^1, 2p_z^1$.

(iii)$1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^2, 3d^3$.

(iv)$1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 3d^5, 4s^1$.

Q23.The first shell may contain up to 2 electrons, the second shell up to 8, the third shell up to 18, and the fourth shell up to 32. Explain this arrangement in terms of the quantum number.

Ans:-For the first shell n=1, l=0,$m_l = 0, m_s$=+1/2and-1/2. It can have 2 electrons both with opposite spins.

For n=2, l=0,$m_l = -1,0,+1$ and m_s =+1/2, -1/2**.**

Therefore, a total of 2+6=8 electrons are present.

For, n=3, l=0,$m_l = 0, m_s$ =+1/2,-1/2,

n=3,l=1, $m_l = -1,0,+1, m_s$ =+1/2,-1/2,

n=3,l=2,$m_l = -2,-1,0, = +1,+2, m_s$ =+1/2,-1/2.

For, n=4,l=0,$m_l = -1,0,+1, m_s = +1/2$,-1/2,

n=4,l=1,$m_l = -2,-1,0,+1,+2, m_s$ =+1/2,-1/2,

n=4,l=2,$m_l = -2,-1,0,+1,+2,+3, m_s$ =+1/2,-1/2,

n=4,l=3,$m_l = -3,-2,-1,0,+1,+2,+3, m_s$=+1/2,-1/2.

Therefore, a total of 2+6+10+14=32 electrons are presents.

Q24. The quantum number of six electrons are given below. Arrange them in order of increasing energies. If any of these combination has/have the same energy lists:

(i)n=4,l=2,$m_l = -2, m_s$ =-1/2

(ii)n=3,l=2,$m_l = 1, m_s$ =+1/2
(iii)n=4,l=1,$m_l = 0, m_s$ =+1/2
(iv)n=3,l=2,$m_l = -2, m_s$ =-1/2
Ans:-(i)4d,(n+l)=6
(ii)3d,(n+l)=5
(iii)4p,(n+l)=5
(iv)3d,(n+l)=5
(ii)=(iv)<(iii)<(i)
Therefore, 3d=3d<4p<4d

Q25. The arrangement of orbitals on the basis of energy is based upon their (n+l) value. Lower the value of (n+l), lower is the energy. For orbitals having same values of (n+l), the orbitals with lower values of n will have lower energy.
Based upon the above information, arrange the following orbitals in the increasing order of energy.
(a)1s,2s,3s,2p (b)4s,3s,3p,4d
(c)5p,4d,5d,4f,6s (d)5f,6d,7s,6p
Ans:-(i)(a)(n+l) values are

1s=1+0=1,2s=2+0=2

3s=3+0=3,2p=2+1=3

Hence, increasing order of their energy is

1s<2s<2p<3s.

(b) 4s=4+0=0,33=3+0=3

3s=3+1=4,4d=4+2=6

Hence, 3s<3p<4s<4d

(c) 5p=5+1=6,4d=4+2=6

5d=5+2=7,4f=4+3=7

6s=6+0=6

Hence, 4d<5p<6s<4f<5d.

(d) 5f=5+3=8,6d=6+2=8

7s=7+0=7,7p=7+1=8

Hence,7s<5f<6d<7p

Q26.Write down the quantum number n, l and m for the following orbitals

(i)$3d_{x^2-y^2}$ (ii)$4d_{z^2}$

(iii)$3d_{xy}$(iv)$4d_{xz}$

Ans(i)n=3,l=2,m=+2

(ii)n=4,l=2,m=0

(iii)n=3,l=2,m=-2

(iv)n=4,l=2,m=+1.